USN | | | | | | | | | | | [22TCSDE23]

SIXTH SEMESTER B.Tech EXAMINATION, FEBRUARY - 2025

INTRODUCTION TO NATURAL LANGUAGE PROCESSING

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

			Blooms Level	CO	Marks
		PART - A			**5 x2=10**
1.	**a.**	Define tokenization and its role in text processing in Natural Language Processing	L1	CO1	**02**
	b.	Show how the Viterbi algorithm is used in Hidden Markov Models (HMM) for POS tagging.	L1	CO2	**02**
	c.	What is Probabilistic Context-Free Grammar (PCFG).	L1	CO3	**02**
	d.	Define Word Sense Disambiguation (WSD) and explain its challenges in NLP.	L1	CO4	**02**
	e.	Explain Latent Dirichlet Allocation (LDA) briefly.	L2	CO5	**02**
		PART - B			**6 X 5=30**
2.	**a.**	Illustrate why is NLP considered hard. Discuss with examples how ambiguity in language (lexical, syntactic, and semantic) contributes to the challenges.	L2	CO1	**05**
		OR			
	b.	Demonstrate how the edit distance method can be used to calculate the distance between the words "intention" and "execution." Explain the steps clearly.	L2	CO1	**05**
3.	**a.**	Explain the finite-state methods used in computational morphology. How do these methods differ from other morphological analysis techniques?	L2	CO2	**05**
		OR			
	b.	Compare and contrast the Viterbi algorithm and the Baum-Welch algorithm in the context of training and decoding in Hidden Markov Models.	L2	CO2	**05**
4.	**a.**	Illustrate dependency grammar and how they differ from constituency grammars. What are the key components of a dependency grammar?	L2	CO3	**05**
		OR			
	b.	Explain the concept of syntactic parsing and tell how it can affect the interpretation of sentences?	L2	CO3	**05**
5.	**a.**	Compare and contrast distributional models of semantics with structured models of semantics.	L2	CO4	**05**

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		OR			
	b.	Using the CKY parsing algorithm, parse the following sentence: "The cat sat on the mat." Demonstrate the steps and the parsing table created during the process.	L2	CO4	**05**
6.	**a.**	Explain word embeddings and their significance in natural language processing. How do they differ from traditional vector space models?	L2	CO5	**05**
		OR			
	b.	Infer about the distributional models of semantics along with the core principle behind the distributional hypothesis.	L2	CO5	**05**
7.	**a.**	Explain Gibbs Sampling as it relates to LDA. How does this technique facilitate the estimation of topic distributions in documents?	L2	CO6	**05**
		OR			
	b.	List out the different approaches of text summarization and discuss the criteria for evaluating the effectiveness of summarization techniques.	L1	CO6	**05**
		PART - C		**6 X 10=60**	
8.	**a.**	Demonstrate how the Noisy Channel Model for spelling correction. Outline how this model can be implemented to correct misspelled words in a text. Include a simple example to demonstrate the process.	L2	CO1	**10**
		OR			
	b.	Discuss the significance of N-Gram language models in NLP. Evaluate the strengths and limitations of N-Gram models compared to other language modeling techniques. How does basic smoothing influence the performance of N-Gram models?	L2	CO1	**10**
9.	**a.**	Consider a scenario where you have limited training data for POS tagging. Analyze the potential challenges this poses for both Hidden Markov Model (HMM) and Maximum Entropy (MaxEnt) model training	L3	CO2	**10**
		OR			
	b.	Illustrate the core principles of Finite State Automata (FSA) and Finite State Transducers (FSTs) as applied to computational morphology. Explain, with examples, how these models are used to represent and process word forms, covering both inflectional and derivational morphology	L2	CO2	**10**
10.	**a.**	Interpret the process of shallow parsing and explain its purpose in NLP. Design a shallow parsing strategy for identifying noun phrases in English text, including the specific techniques and types of features you would employ.	L2	CO3	**10**
		OR			
	b.	Demonstrate how the CKY algorithm can be used to parse a sentence given a specific Context-Free Grammar (CFG) in Chomsky Normal Form (CNF). Provide a sample sentence, a small CNF grammar, and show all steps of the CKY algorithm applied to this example.	L2	CO3	**10**

11.	**a.**	Explain the process of generating word embeddings using a neural network model such as Word2Vec (or similar). Illustrate with an example how these embeddings capture semantic relationships between words.	L2	CO4	**10**
		OR			
	b.	Identify the role of WordNet in lexical semantics. Discuss its advantages as a resource for representing lexical relations (e.g., hypernymy, hyponymy, synonymy).	L3	CO4	**10**
12.	**a.**	Compare and contrast the Earley, and Chart parsing algorithms in the context of natural language processing. List out their relative strengths and weaknesses in handling different types of grammars.	L2	CO5	**10**
		OR			
	b.	Analyze the strengths and limitations of using distributional semantic models, specifically word embeddings, for query expansion in information retrieval. Compare this approach with traditional techniques based on lexical resources.	L3	CO5	**10**
13.	**a.**	Examine the different methods for evaluating the quality of text summarization. Discuss the strengths and weaknesses of each evaluation method and the kind of aspects of the summary they measure and justify which method is most suitable for different summarization tasks.	L4	CO6	**10**
		OR			
	b.	Illustrate, with a simple example, how LDA might group documents based on their underlying topic structure. List out the key assumptions behind LDA.	L2	CO6	**10**